

Lab #6

PIC18F Conditional I/O (polled I/O) Using Assembly and C

ECE 3301L Introduction to Microcontrollers Laboratory

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Videos

Prelab:

Code: https://youtu.be/fxrfMHB-_T0?si=Il6tlv-qDDj_ssvJ

Circuit: <https://youtu.be/oVuvvg5KTi70?si=BaISqjz17BQkFR7w>

Postlab:

Code and circuit: <https://youtu.be/7LUBXifRsGE?si=0hnB18LRRDyfgDr3>

Objective: The purpose of this lab is to illustrate the concept of PIC18F4321 conditional or polled I/O.

PRELAB:

prelab i)

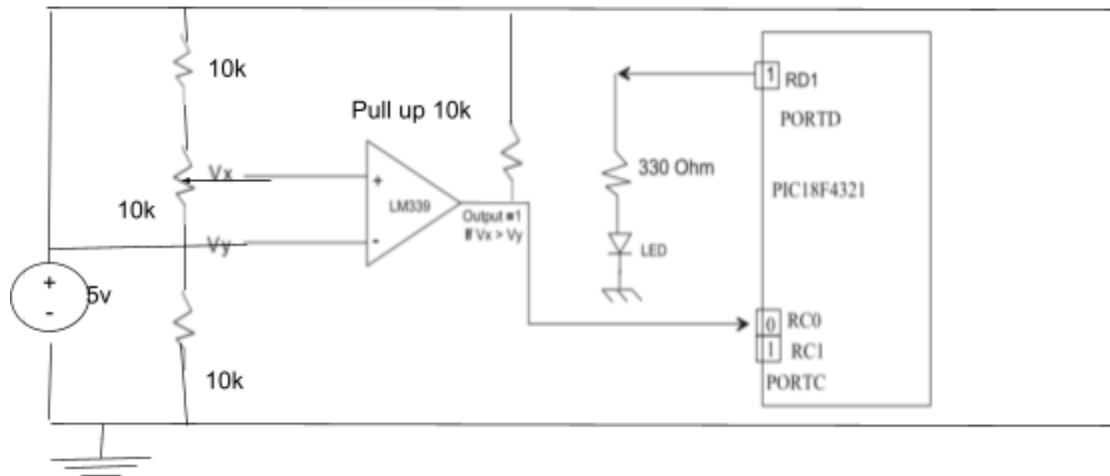
In the above figure , the PIC18F4321 is required to turn on an LED connected to bit 1 of PORTD if the comparator voltage $V_x > V_y$; otherwise, the LED will be turned off. Write a PIC18F assembly language program at address 0x200 to accomplish this using:

- (a) Conditional or Polled I/O
- (b) Repeat (a) using C

Equipment, Software and Components required:

- 10k Ω potentiometer
- 10k Ω resistors 3x
- 1k Ω resistor
- 330 Ω resistor
- LED (2-2.2 voltage drop)
- Push button
- MPLAB IDE 5.40
- XC8 2.20
- PIC18F4321
- PicKit3
- Breadboard
- LM339N comparator
- Power supply
- Wires and leads with alligator clips

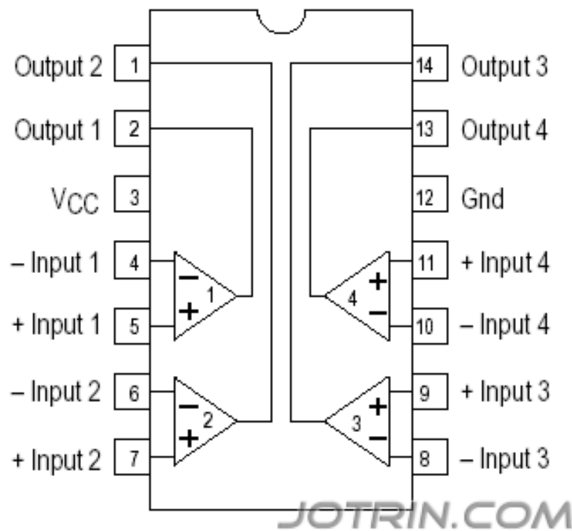
Explanation of hardware:



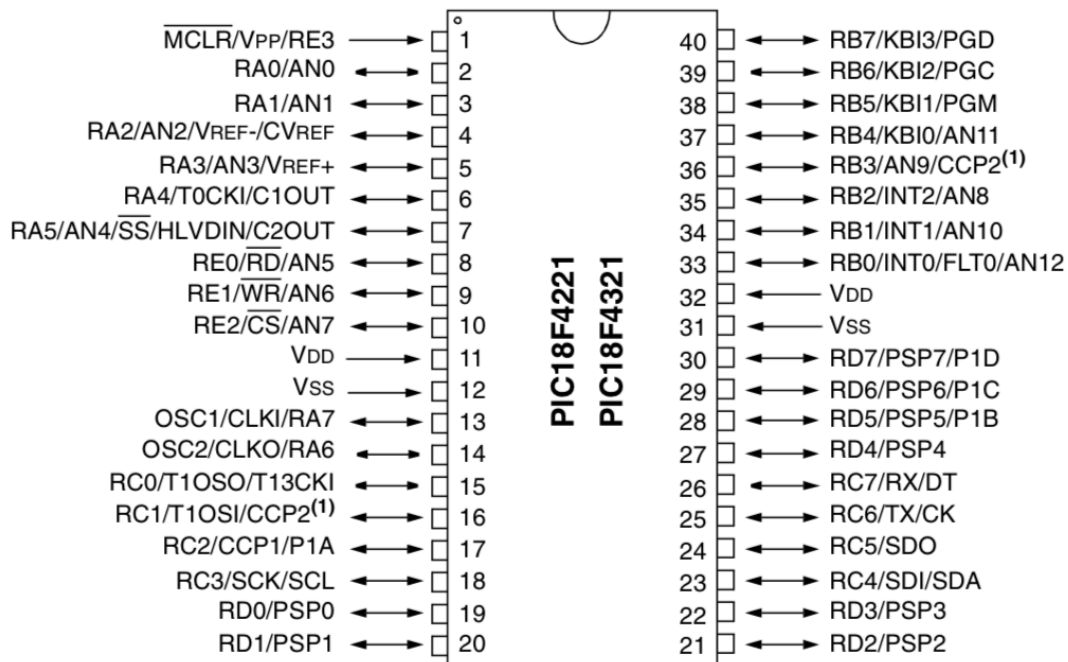
IMG 1. *The schematic of the circuit*

The 5v serves as V_y , and is also the power for pins 11 and 32 of the PIC18F, and pin 3 of the LM339n. Everything is connected to the same ground. In this manner, only one voltage source needs to be used to have two different voltage inputs to compare as well as to provide power to the Pic and comparator. Note that the PIC was pre-programmed before going into the circuit, so connections to the pickit are unnecessary.

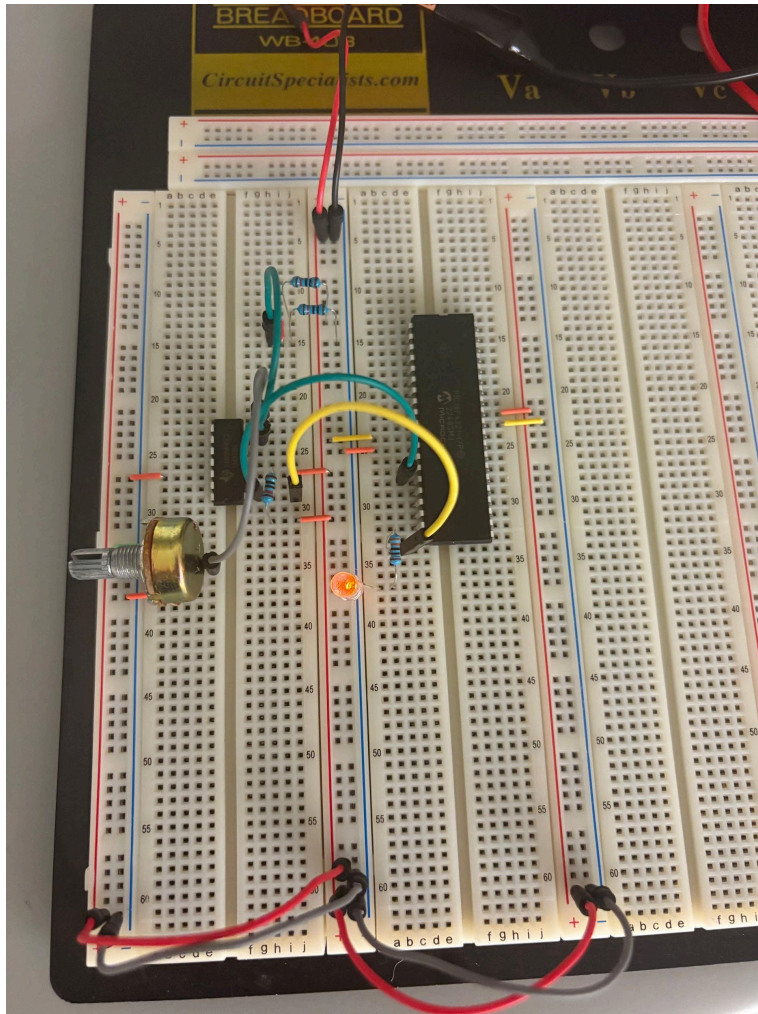
LM339N Pinout



IMG 2. Layout of the comparator, found at jotrin.com. More information is at the end of this report.



IMG 3. Layout of the PIC18F4321, from microchip.com. More information is at the end of this report.

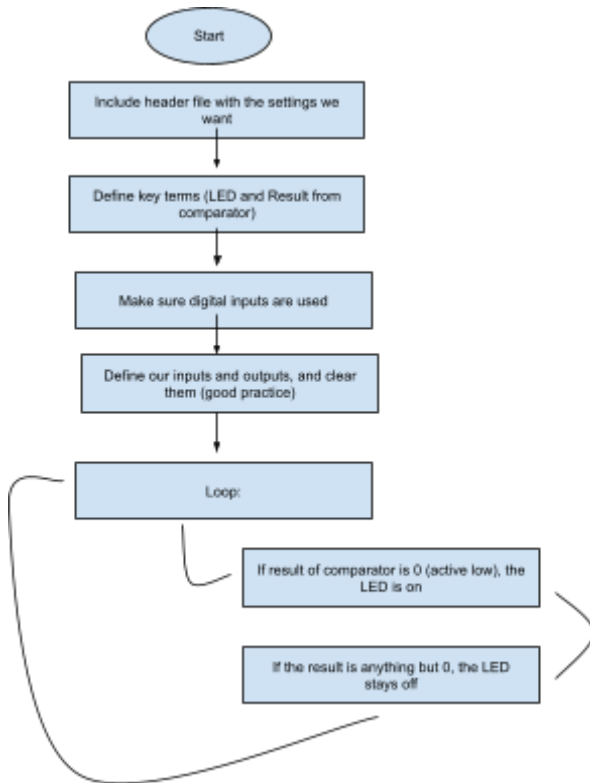


IMG 4. *The physical circuit.*

Code:

1. Set up configuration -
 - OSC= INTIO2
 - BOR=OFF
 - WDT = OFF
 - STVREN = OFF
 - MCLRE = ON
 - LVP=OFF
2. Initialize at 0x200
3. Set all analog pins to be digital
4. Set port C to be an input, and clear it
5. Set port D to be an output, and clear it
6. Make a loop for repeatedly checking the outputs-
 1. Test bit 0 of port C, if it is 0 turn on the LED
 2. Otherwise, loop continues

POSTLAB:



IMG 5. *Flow chart*

What are the basic differences between conditional I/O and interrupt I/O?

Conditional (or polling) I/O involves continuous checking of the input, while for interrupt I/O, the device that is the input “tells” the CPU to check the input. For conditional, the CPU program samples a status signal periodically and takes actions in response to the state read. Each I/O device is checked by a continuous polling cycle. For interrupt, when the status pin changes state, the CPU is prompted to call an interrupt routine to take care of the I/O. Basically, external input is needed to tell the processor that there is an input needing to be addressed, so the

processor does not need to repeatedly and automatically check the inputs. Constantly checking the input for conditional I/O can be inefficient and strain the CPU, but interrupt I/O is a bit more complicated to use.

Summarize the basic differences between PIC18F external and internal interrupts

External interrupts are initiated by outside devices and hardware, and received in the ports of the microcontroller. This is essentially communication between hardware and software. Internal interrupts are interrupts activated by internal components of the microcontroller (like timers) or by errors in the program execution.

WORKS CITED

- *Microchip*, ww1.microchip.com/downloads/en/DeviceDoc/39689b.pdf. Accessed July 2025.
- [Www.jotrin.com](http://www.jotrin.com), Jotrin |. "LM339N Comparator Datasheet, Pinout, and Circuit." *Jotrin Electronics*, www.jotrin.com/technology/details/lm339n-datasheet-circuit-and-pinout. Accessed July 2025.